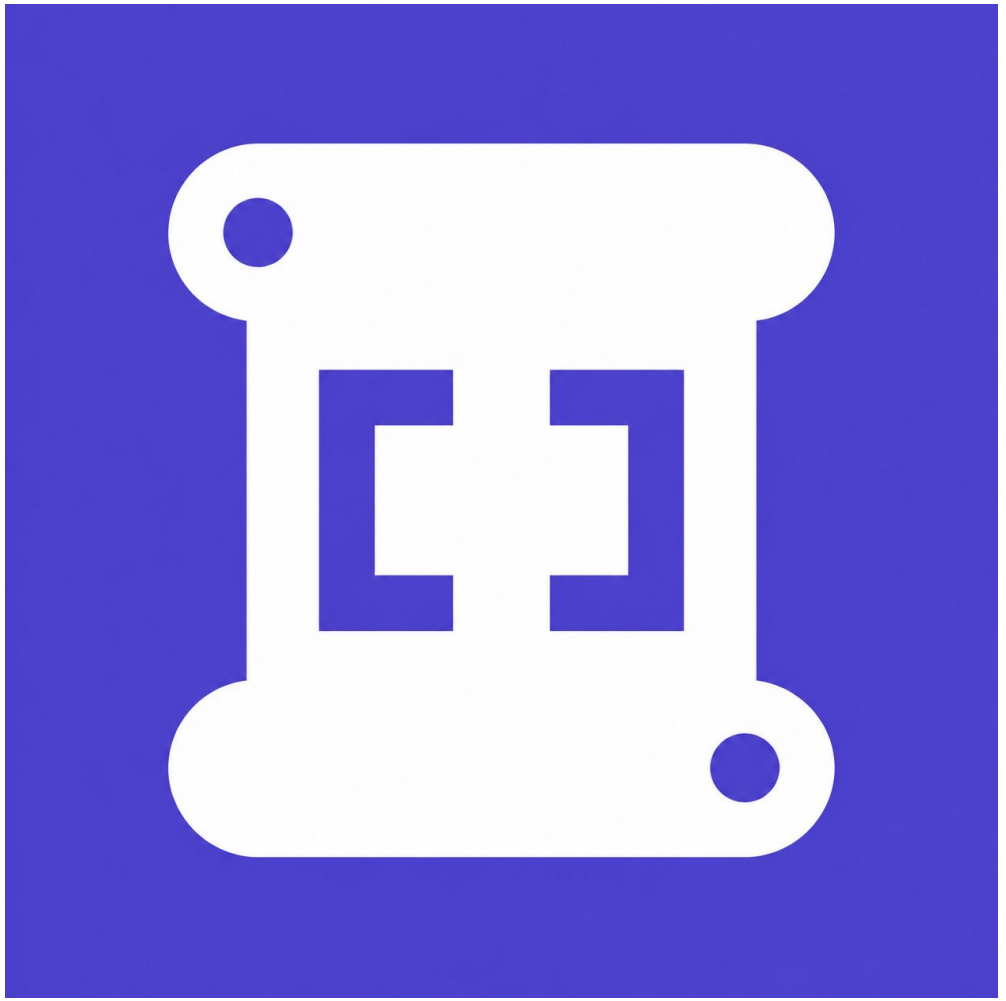
NeoPapyrus

RESEARCH PAPER DISCOVERY ASSISTANT

Plenty of tools summarize papers. *NeoPapyrus makes them debate.*

Uncovering critical research gaps that nobody has filled yet 100% inside Snowflake.



Pramit Patel · Rohan Karve · Carlos Mata · Chance Caswell

BUILT ON SNOWFLAKE

Literature reviews take weeks.

Summarizing isn't the hard part finding where papers disagree is.

Hidden Conflicts

Researchers can't easily find contradictions or unanswered questions across dozens of research topics, including medical research

935+

Selection Noise

It's impossible to tell quickly which papers are worth reading in-depth versus skipping entirely.

WORTH READING?

Hallucinated AI

Generic chatbots make up non-existent citations and fake references.

Zero Trust

Four steps to absolute research clarity

17 real research papers · 935 searchable passages

Ask

Exact passage citations + "Should you read it?" verdict (Must Read / Related / Skip).

★ FLAGSHIP FEATURE

Paper Debate

Pick two papers. Each side argues using *only* its own text, then an AI judge rules on who has stronger evidence.

ADVERSARIAL AI ANALYSIS

Research Gaps

Uncovers consensus, disagreements, and formulates 3 open research questions.

Insights

Analyzes questions to surface top-cited papers and trending healthcare topics.

How It Works

Everything runs inside one Snowflake account, zero outside servers.

01
PDF Stage
Snowflake internal stage

02
AI Parse
AI_PARSE_DOCUMENT

03
Embed
Chunks + AI_EMBED

04
Search & LLM
Vector similarity & Llama 3.3

05
Streamlit
Native App UI

⚡ **Also accessible via Snowflake's Cortex REST API from any external system.**

"Take Snowflake away and nothing is left."

A complete technical stack built across 10 native Snowflake features.

Internal stage + directory table

AI_PARSE_DOCUMENT

Text chunking algorithms

AI_EMBED + VECTOR type

VECTOR_COSINE_ SIMILARITY

AI_COMPLETE (Llama 3.3 70B)

VARIANT, ARRAY, FLATTEN

Streamlit in Snowflake

Snowpark Python

Cortex REST API

THE STAR OF THE SHOW

Live Demo

Ask → Debate → Research gaps

Watch NeoPapyrus synthesize complex AI healthcare literature in real time.

What's Next

Expanding from hackathon prototype to full researcher platform.

- Automatic arXiv & PubMed paper imports
- Visual agreement & disagreement consensus map
- Export draft literature reviews with citations
- Student comprehension quiz mode

Thank you. Questions?

Built by Pramit Patel, Rohan Karve, Carlos Mata, & Chance Caswell